Questions and MCQ
Step 2 : Remind Value Creation Investment Decision Rules

1) Which of the following statements is FALSE?
A) About 75% of firms surveyed used the NPV rule for making investment decisions.
B) If you are unsure of your cost of capital estimate, it is important to determine how sensitive your analysis is to errors in this estimate.
C) To decide whether to invest using the NPV rule, we need to know the cost of capital.
D) NPV is positive only for discount rates greater than the internal rate of return.
Answer: D

Use the following information to answer the question(s) below.

Sarah Palin reportedly was paid a $11 million advance to write her book Going Rogue. The book took one year to write. In the time she spent writing, Palin could have been paid to give speeches and appear on TV news as a political commentator. Given her popularity, assume that she could have earned $8 million over the year (paid at the end of the year) she spent writing the book.

2) Assuming that Palin's cost of capital is 10%, then the NPV of her book deal is closest to:
A) $2.00 million.
B) $2.20 million.
C) $3.00 million.
D) $3.73 million.
Answer: D
Explanation: NPV = $11 million - $8 million/(1.10)1 = $ 3.72727 million

Use the information for the question(s) below.

The Sisyphean Company is planning on investing in a new project. This will involve the purchase of some new machinery costing $450,000. The Sisyphean Company expects cash inflows from this project as detailed below:

Year One	Year Two	Year Three	Year Four
$200,000	$225,000	$275,000	$200,000

The appropriate discount rate for this project is 16%.

3) The NPV for this project is closest to:
A) $176,270.
B) $123,420.
C) $450,000.
D) $179,590.
Answer: A

Explanation: NPV = -$450,000+ $200,000/(1.16)1 + $225,000/(1.15)2 + $275,000/(1.15)3 + $200,000/(1.15)4 = $176,265

Use the following information to answer the question(s) below.

Frank Dewey Esquire from the firm of Dewey, Cheatum, and Howe, has been offered an upfront retainer of $30,000 to provide legal services over the next 12 months to Taggart Transcontinental. In return for this upfront payment, Taggart Transcontinental would have access to 8 hours of legal services from Frank for each of the next 12 months. Frank's normal billable rate is $250 per hour for legal services.

4) Assuming that Dewey's cost of capital is 12% EAR, then the NPV of his retainer offer is closest to:
A) -$7500.
B) -$7400.
C) $6000.
D) $7400.
Answer: D

Explanation: Step #1: Monthly Interest Rate $\sqrt[12]{1 + .12} - 1 = .009488793$
Step #2: Monthly Opportunity Cost = 8 × $250 = $2000
Step #3: NPV = $30,000 - $2000 $\left[\frac{1}{.009488793}\right] - \left[\frac{1}{.009488793(1 + .009488793)^{12}}\right] = \$7416.97$

5) Which of the following statements is FALSE?
A) The IRR investment rule will identify the correct decision in many, but not all, situations.
B) By setting the NPV equal to zero and solving for r, we find the IRR.
C) If you are unsure of your cost of capital estimate, it is important to determine how sensitive your analysis is to errors in this estimate.
D) The simplest, though least accurate, investment rule is the NPV investment rule.
Answer: D

Use the information for the question(s) below.

The Sisyphean Company is planning on investing in a new project. This will involve the purchase of some new machinery costing $450,000. The Sisyphean Company expects cash inflows from this project as detailed below:

Year One	Year Two	Year Three	Year Four
$200,000	$225,000	$275,000	$200,000

The appropriate discount rate for this project is 16%.

6) The IRR for this project is closest to:
A) 18.9%.
B) 22.7%.
C) 34.1%.
D) 39.1%.

Answer: C
Explanation: CF0 = -450,000
CF1 = 200,000
CF2 = 225,000
CF3 = 275,000
CF4 = 200,000
Compute IRR = 34.12%

Use the following information to answer the question(s) below.

Rearden Metals is considering opening a strip-mining operation to provide some of the raw materials needed in producing Rearden metal. The initial purchase of the land and the associated costs of opening up mining operations will cost $100 million today. The mine is expected to generate $16 million worth of ore per year for the next 12 years. At the end of the 12th year Rearden will need to spend $20 million to restore the land to its original pristine nature appearance.

7) The payback period for Rearden's mining operation is closest to:
A) 5.00 years.
B) 6.00 years.
C) 6.25 years.
D) 6.50 years.
Answer: C
Explanation: Payback = $100 million/$16 million per year = 6.25 years

8) Which of the following statements is FALSE?
A) The payback rule is useful in cases where the cost of making an incorrect decision might not be large enough to justify the time required for calculating the NPV.
B) The payback rule is reliable because it considers the time value of money and depends on the cost of capital.
C) For most investment opportunities, expenses occur initially and cash is received later.
D) Over fifty percent of firms surveyed reported using the payback rule for making decisions.
Answer: B
Explanation: The payback rule does not take into account the time value of money.

Use the table for the question(s) below.

Consider the following two projects with cash flows in $:

Project	Year 0 Cash Flow	Year 1 Cash Flow	Year 2 Cash Flow	Year 3 Cash Flow	Year 4 Cash Flow	Discount Rate
A	-100	40	50	60	N/A	.15
B	-73	30	30	30	30	.15

9) Assume that projects A and B are mutually exclusive. The correct investment decision and the best rationale for that decision is to:
A) invest in project A since NPVB < NPVA.

B) invest in project B since IRRB > IRRA.
C) invest in project B since NPVB > NPVA.
D) invest in project A since NPVA > 0.
Answer: C
Explanation: NPVA = -$100 + $40/(1.15)1 + $50/(1.15)2 + $60/(1.15)3 = $12.04
NPVB = -$73 + $30/(1.15)1 + $30/(1.15)2 + $30/(1.15)3 + $30/(1.15)4 = $12.64

IRR A
CF0 = -100
CF1 = 40
CF2 = 50
CF3 = 60
Compute IRR = 21.65%

IRR B
CF0 = -73
CF1 = 30
CF2 = 30
CF3 = 30
CF4 = 30
Compute IRR = 23.34%

10) Which of the following statements is FALSE?
A) If there is a fixed supply of a resource available, you should rank projects by the profitability index, selecting the project with the lowest profitability index first and working your way down the list until the resource is consumed.
B) Practitioners often use the profitability index to identify the optimal combination of projects when there is a fixed supply of resources.
C) If there is a fixed supply of resources available, so that you cannot undertake all possible opportunities, then simply picking the highest NPV opportunity might not lead to the best decision.
D) The profitability index is calculated as the NPV divided by the resources consumed by the project.
Answer: A

Use the information for the question(s) below.

The Sisyphean Company is planning on investing in a new project. This will involve the purchase of some new machinery costing $450,000. The Sisyphean Company expects cash inflows from this project as detailed below:

Year One	Year Two	Year Three	Year Four
$200,000	$225,000	$275,000	$200,000

The appropriate discount rate for this project is 16%.

11) The profitability index for this project is closest to:
A) .44.
B) .26.

C) 0.39.
D) .34.
Answer: C
Explanation: PI = NPV/Investment

NPV = -$450,000 + $200,000/(1.16)1 + $225,000/(1.16)2 + $275,000/(1.16)3 + $2,000,000/(1.16)4 = $176,265

So, PI = $176,265/$450,000 = 0.39